

EDUCATION

Columbia University

NEW YORK

B.A candidate in Computer Science and Mathematics. GPA: 3.93/4.00

AUG 2022 - PRESENT

Selected Coursework: COMS 6998 Theory of Large Language Models, COMS 4774 Unsupervised Learning, COMS 4771 Machine Learning, COMS4731 Computer Vision, CSOR4231W Analysis of Algorithms

Raffles Institution

SINGAPORE

Singapore-Cambridge GCE A-Levels: 6 Distinctions

JAN 2014 - DEC 2019

- Coursework: Physics(H2 and H3), Chemistry, Math, Economics, General Paper
- Enrolled in Raffles Academy, a highly competitive and selective advanced STEM program

Science Society - Vice Chairman

- Oversaw the planning and organization of events such as the Science Olympiad held by the school, Science Advocacy week, speaker events etc.. Prepared weekly club sessions for club members

SELECTED AWARDS AND ACHIEVEMENTS

International Olympiad on Astronomy and Astrophysics: Bronze Medal

AUG 2019

- Achieved a gold medal and an individual 5th ranking in the National Astronomy Olympiad. Represented Singapore at the International Olympiad on Astronomy and Astrophysics and achieved a bronze medal

Singapore Space Challenge: Overall Champions

JUN 2019

- Designed and modeled a CubeSatellite for use in the detection of wildfires. Awarded the grand prize and a monetary prize of SGD10,000

Singapore Physics Olympiad: Silver Medal

NOV 2018

- Achieved a silver medal in the National Physics Olympiad

MIT Pokerbots 2024: Rank 11th/96 Teams

JAN 2024

- Built a Python bot that played a variant of heads-up poker against opponent bots; overall win rate of 73%

QUANTIACS Q18 Algorithmic Trading Contest: Rank 9th

MAR 2024

- Coded, back-tested and live-tested several algorithms for trading stocks using Python

PROFESSIONAL AND RESEARCH EXPERIENCE

Complex Resilient Intelligent Systems Laboratory (CRIS Lab): Research Assistant

NEW YORK

- Conducted research on the mechanisms behind self-organization and the emergence of collective intelligence in deep neural networks

AUG 2023 - MAY 2024

Luce Maintenance Group Singapore: Software Engineering Intern

SINGAPORE

- Developed and deployed end-to-end features, including frontend UI components using React and backend APIs with Node.js, to enhance user experience
- Worked on optimizing codebase to reduce computational costs and execution speeds for Ruby on Rails web service app. Enhanced database and querying efficiency, resulting in a 30% reduction in server loads

JUN 2023 - AUG 2023

AlQuraishi Lab at Columbia University: Research Assistant

NEW YORK

- Investigated the reimplementation of AlphaFold2 (a machine learning framework developed for predicting protein structure) to create a memory efficient and trainable method for predicting intermolecular interactions

MAR 2023 - AUG 2023

Victoris: Co-Founder

SINGAPORE

- Co-Founder of an education group providing academic enrichment programmes to over 150 students
- Organized talks and information sessions for disadvantaged children, including topics such as College Admissions and Information Access

JUN 2021 - AUG 2022

Republic of Singapore Air Force: Immediate Response Force (mandatory military service)

SINGAPORE

- Performed security operations at Paya Lebar Airbase as a member of the Immediate Response Force under the Perimeter Protection Flight

JAN 2020 - NOV 2021

Defence Science Organization: Research Intern

SINGAPORE

- Designed and created numerical models of reflectarray antennas using COMSOL Multiphysics, Python, MATLAB and AutoCad that achieved perfect agreement with experimental data
- Worked with PCB fabrication and 3D printing technology to construct physical models of the antennas for measurement and testing in the Electromagnetics Laboratory

MAY 2018 - JAN 2019

SKILLS

Proficient: Python, Mathematica. **Intermediate:** C, C++, Java, Javascript, HTML, Ruby, MATLAB, COMSOL, Microsoft Excel, AutoCAD, PostgreSQL

Languages: Native proficiency in English and Chinese. Limited working proficiency in Japanese.